

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – VII

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AIM: Creating Interactive Dashboards and Data Stories.

Part A:- Design a basic dashboard layout using sheets

A1. Open the Dataset

1. Open **Tableau Desktop**.
2. Click **Text File** or the appropriate connection option.
3. Select your IPL dataset.
4. Verify that the required fields are available:
 - Season
 - Winner
 - Venue
 - Toss Decision
 - Match Type
 - Target Runs
 - City
5. Click **Sheet 1** to begin creating visualizations.

A2. Create the Winner Distribution Chart

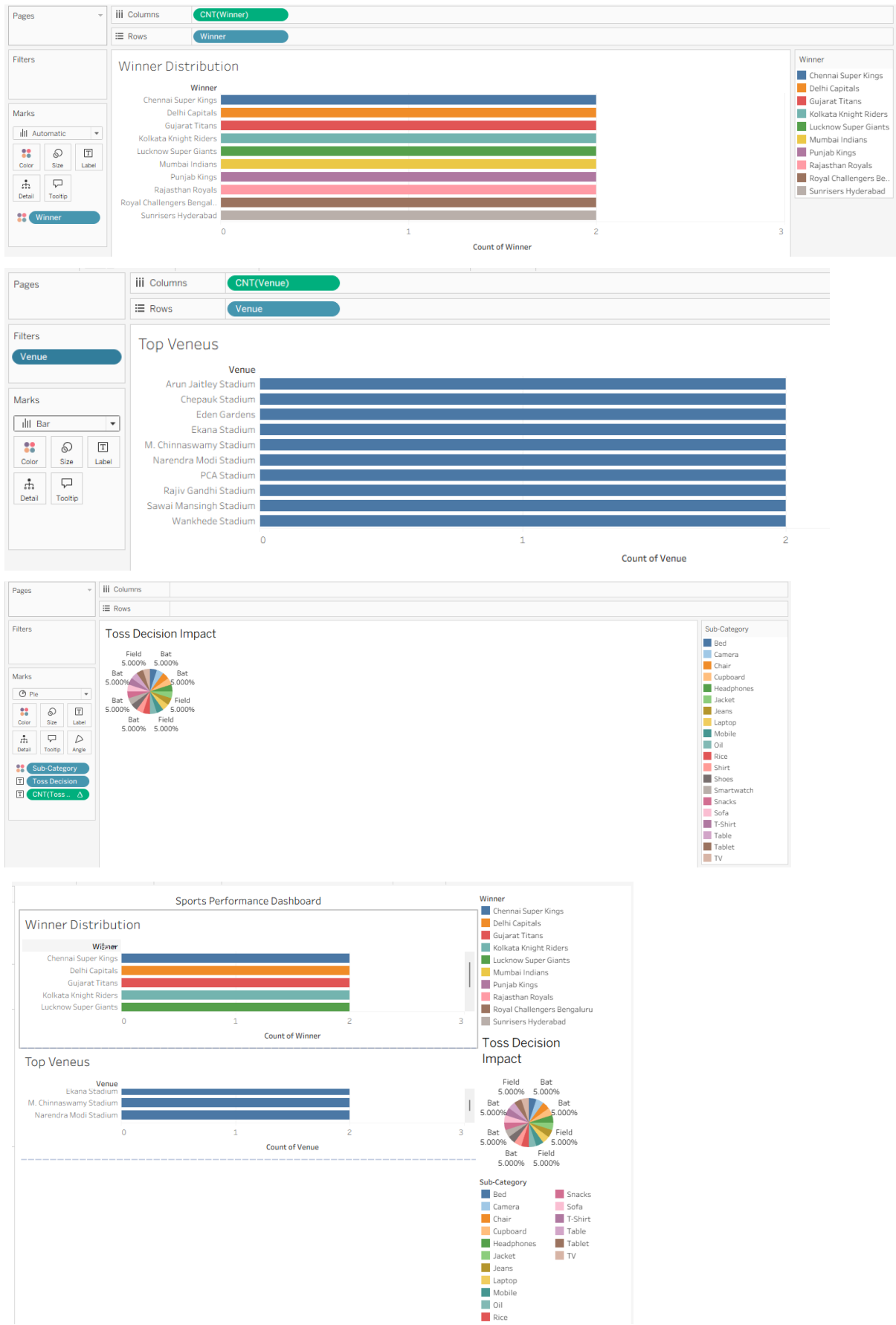
1. Create a new worksheet.
2. Rename it:

Winner Distribution

3. Drag **Winner** to the **Rows** shelf.
4. Drag **Winner** again to the **Columns** shelf.
5. Tableau will automatically create a count such as **CNT(Winner)**.
6. Click **Show Me**.
7. Select **Bar Chart**.
8. Sort the bars in descending order using the sort button.
9. Drag **Winner** to **Color** if you want different colors for each team.
10. Turn on labels:
 - Click **Label** on the Marks card.
 - Tick **Show mark labels**.

Result

The chart displays the number of matches won by each team.



Part B:- Add filters and interactive controls

B1. Create the Worksheet

1. Click **New Worksheet**.
2. Rename it:

Top Venues

3. Drag **Venue** to **Rows**.
4. Drag **Venue** to **Columns**.
5. Tableau creates **CNT(Venue)**.
6. Select **Bar Chart** from Show Me.
7. Sort the bars from highest to lowest.
8. Drag **Venue** to **Label** if necessary.
9. Enable **Show mark labels**.

B2. Limit the Chart to Top Venues

1. Right-click **Venue** on the Rows shelf.
2. Select **Filter**.
3. Click the **Top** tab.
4. Select:

By field

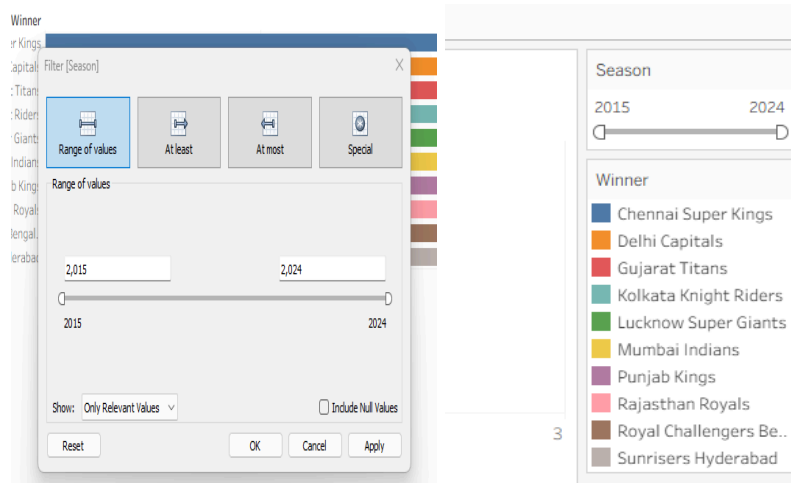
5. Select:

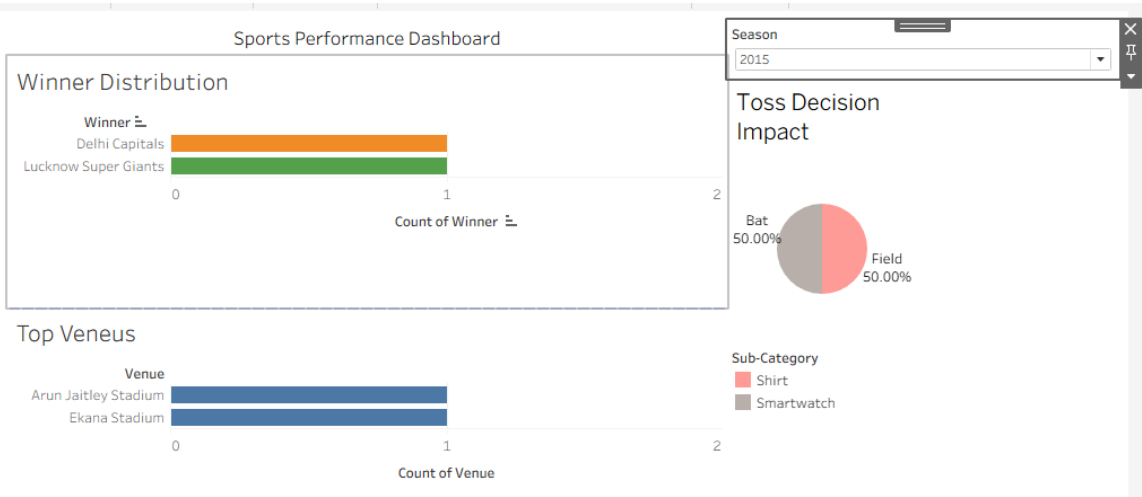
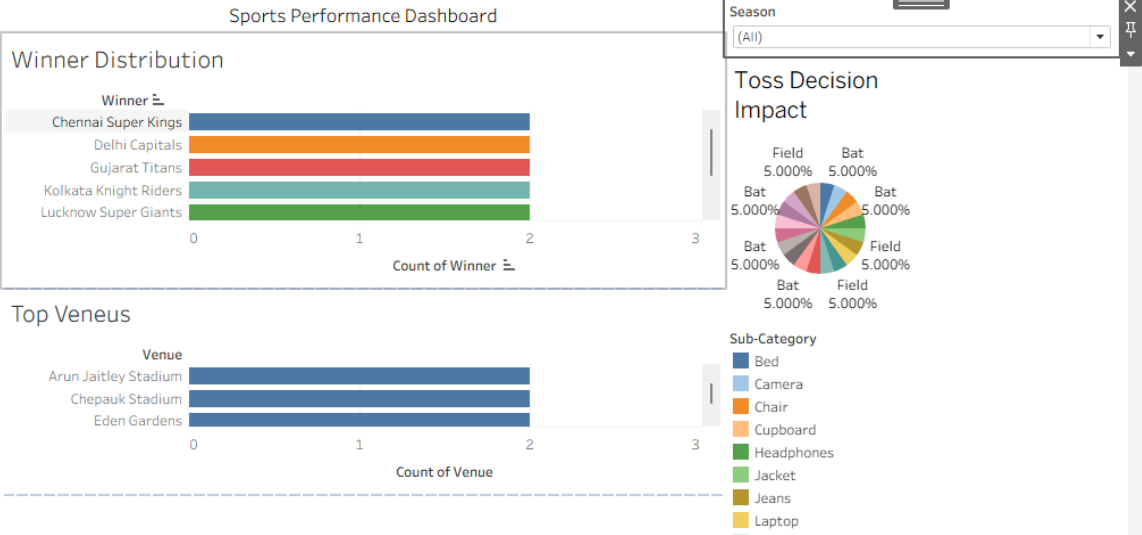
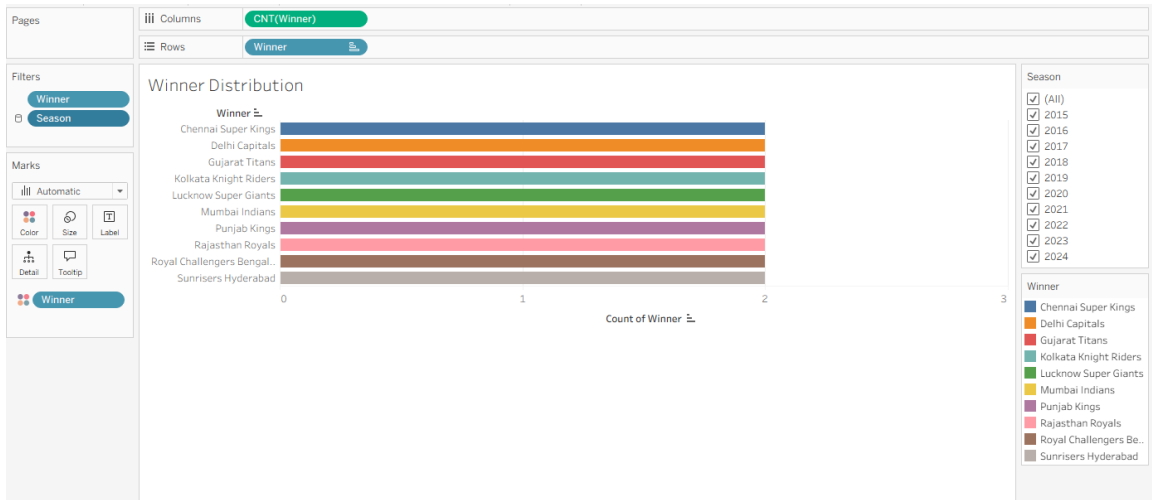
Top 10

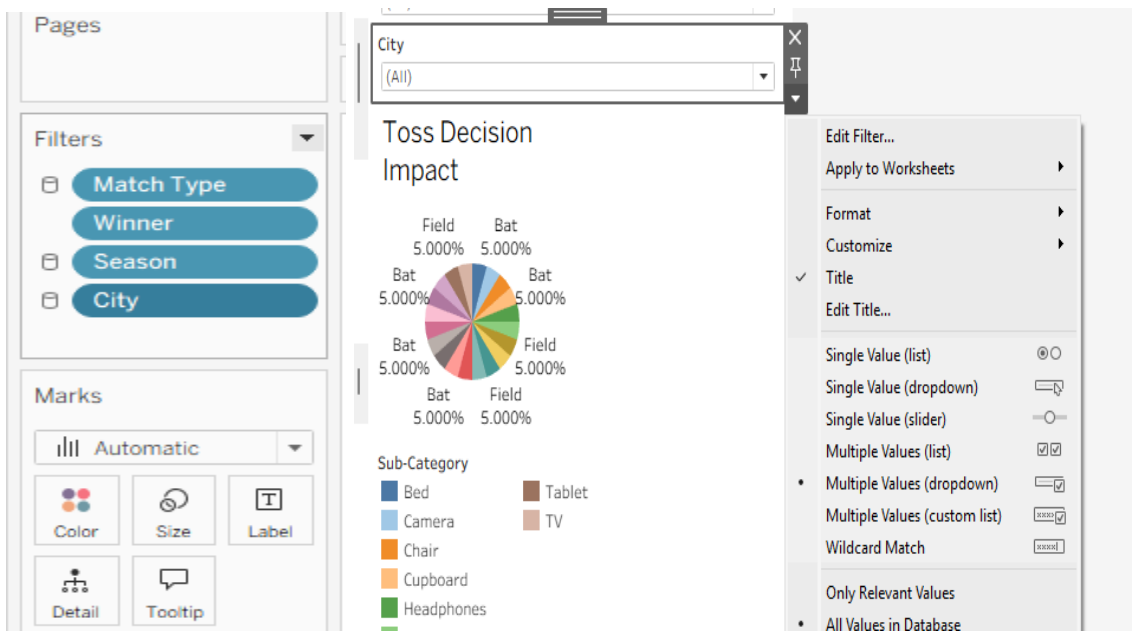
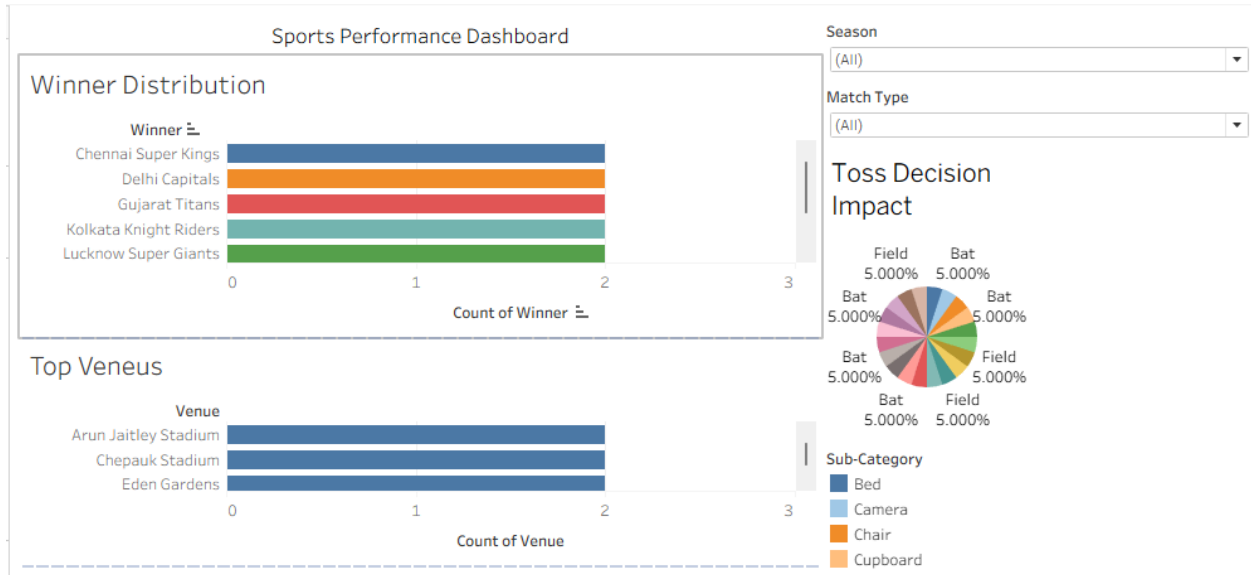
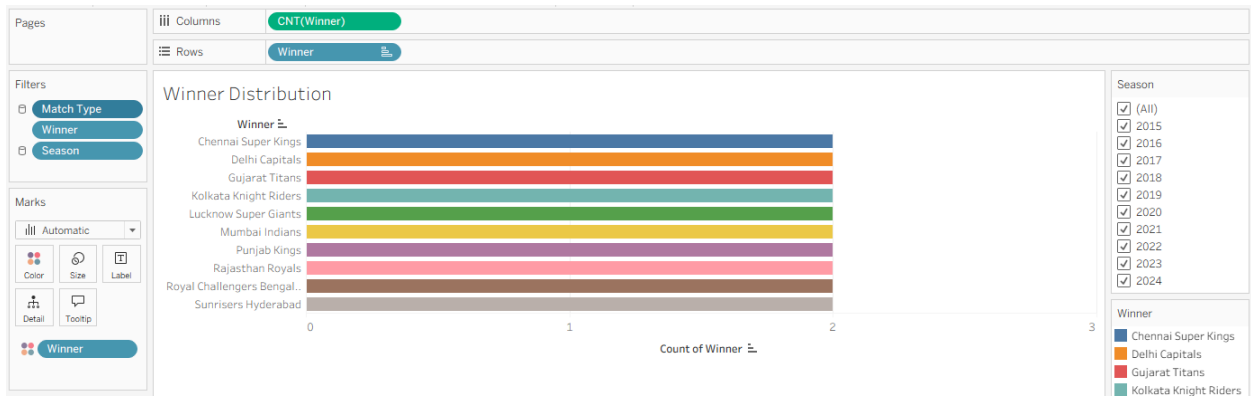
6. Choose:
 - **Count**
 - **Venue**
7. Click **OK**.

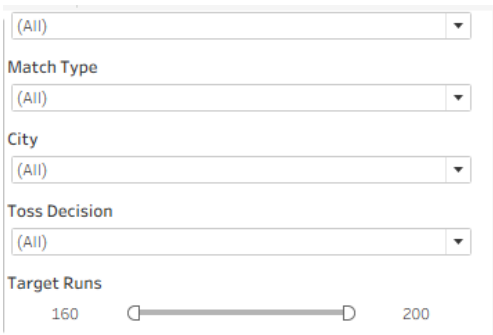
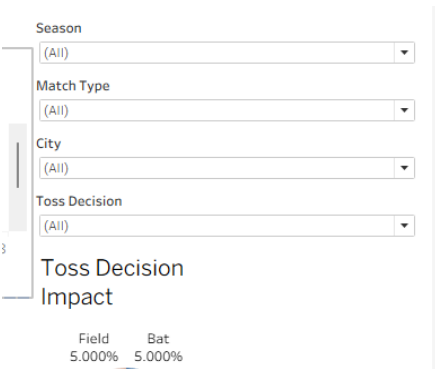
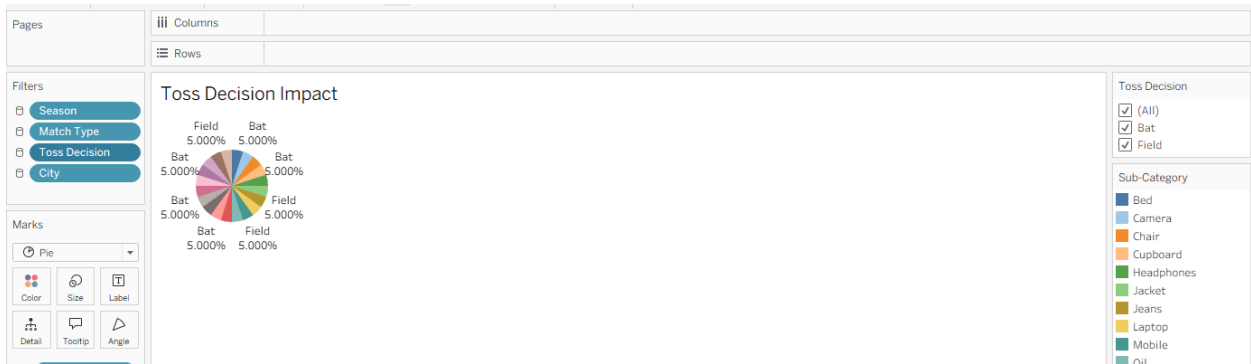
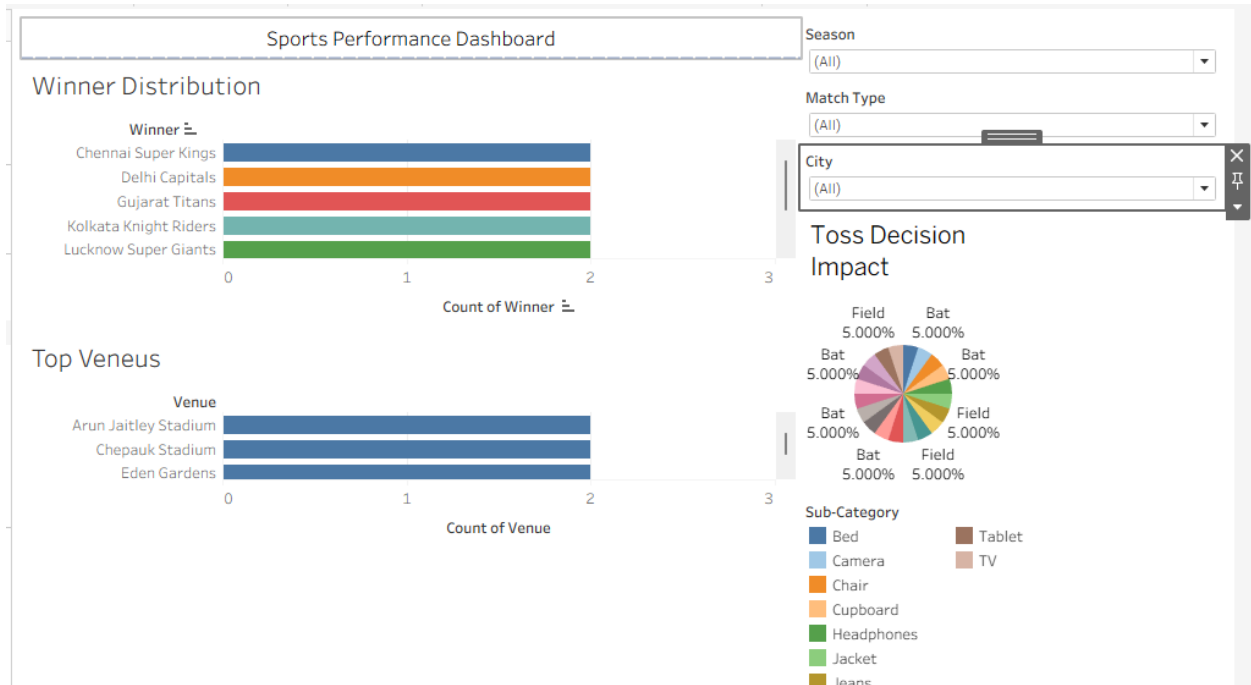
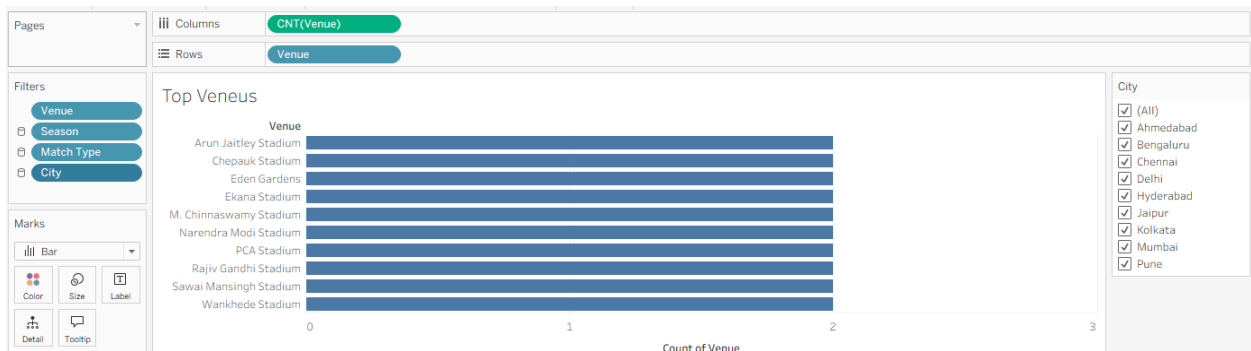
Result

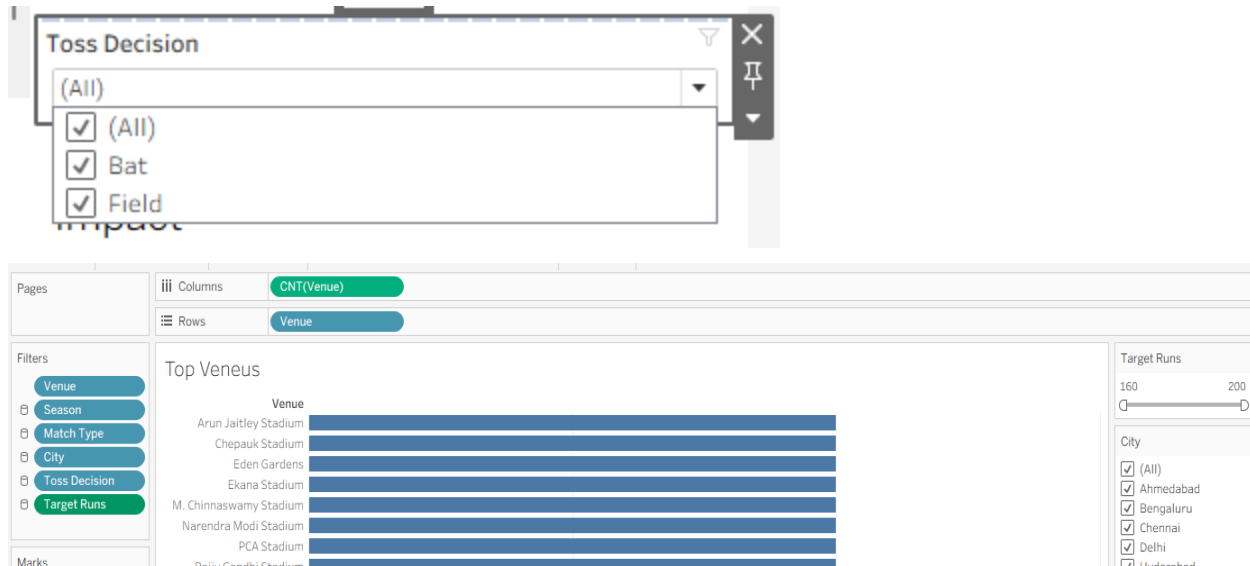
Only the **Top 10 venues** appear.











Part C:- Implement highlight and filter actions

Part C: Creating Filters and Dashboard Interactivity

Your dashboard should use the fields you are actually working with:

- **Season**
- **Match Type**
- **City**
- **Toss Decision**
- **Target Runs**

C1. Add Season Filter

1. Open the worksheet.
2. Drag **Season** to the **Filters** shelf.
3. Select the required seasons or select **All**.
4. Click **OK**.
5. Right-click **Season** on the Filters shelf.
6. Select:

Show Filter

7. The Season filter appears on the right side.

Set it as a dropdown

1. Click the small ▼ on the Season filter card.
2. Select:

Single Value (Dropdown)

Now the user can select one season at a time.

C2. Add Match Type Filter

1. Drag **Match Type** to the **Filters** shelf.
2. Click **OK**.

3. Right-click **Match Type**.
4. Select:

Show Filter

5. Click the filter's ▼.
6. Select:

Single Value (Dropdown)**C3. Add City Filter**

1. Drag **City** to **Filters**.
2. Click **OK**.
3. Select **Show Filter**.
4. Set it to:

Single Value (Dropdown)**C4. Add Toss Decision Filter**

1. Drag **Toss Decision** to **Filters**.
2. Click **OK**.
3. Select **Show Filter**.
4. Select:

Single Value (Dropdown)**C5. Add Target Runs Filter**

1. Drag **Target Runs** to **Filters**.
2. Select the appropriate filtering option.
3. Click **OK**.
4. Right-click **Target Runs**.
5. Select **Show Filter**.
6. On the filter card, select:

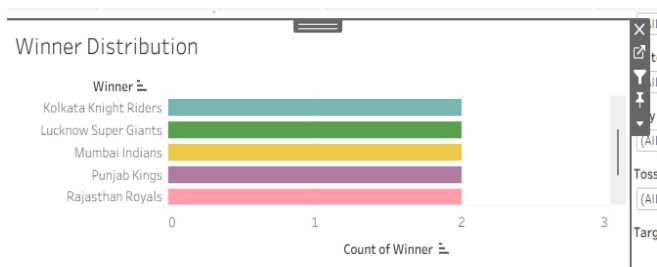
Range of values

A slider will appear.

Result

You can now control the dashboard using:

Season → Match Type → City → Toss Decision → Target Runs



Add Highlight Action

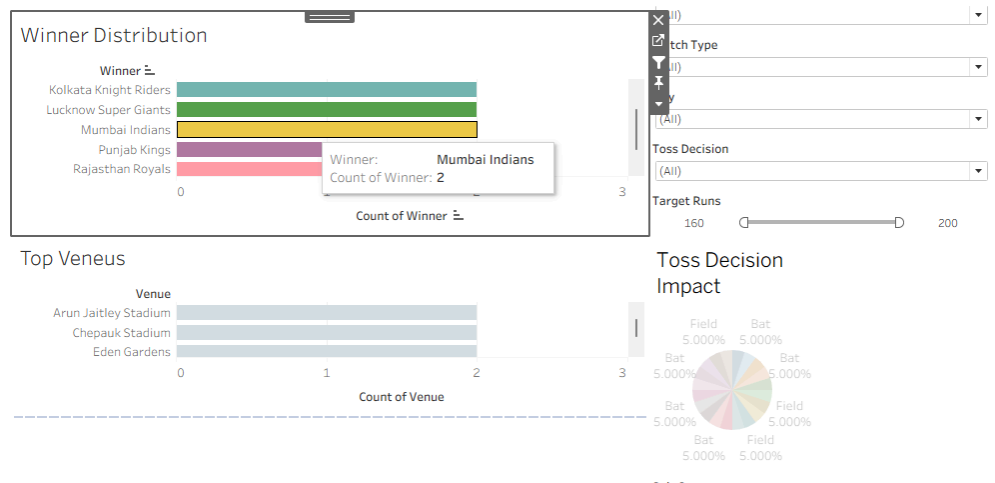
Name: Highlight1

Source Sheets: Dashboard 1 (Sheet 1, Sheet 2, Sheet 3)

Target Sheets: Dashboard 1 (Sheet 1, Sheet 2, Sheet 3)

Target Highlighting: All Fields

Buttons: Cancel, OK



Part D:- Use URL and set actions for interactivity

D1. Create a Set

We will create a set using **Winner**.

1. Open **Winner Distribution**.
2. Find **Winner** in the Data pane.
3. Right-click **Winner**.
4. Select:

Create → Set

5. The Create Set window opens.
6. Name the set:

Selected Teams Set

7. Select one or more teams.
8. Click **OK**.

You will now find **Selected Teams Set** under the **Sets** section.

D2. Use the Set in the Visualization

1. Drag **Selected Teams Set** to **Color** on the Marks card.
2. Tableau will distinguish the teams belonging to the set from those outside the set.

D3. Create a Set Action

1. Open your dashboard.
2. Go to:

Dashboard → Actions

3. Click:

Add Action → Change Set Values

4. Enter:

Update Selected Team Set

5. Under **Source Sheets**, select:

Winner Distribution

6. Under **Run action on**, select:

Select

7. Under **Target Set**, choose:

Selected Teams Set

8. Choose the option to assign the selected values to the set.
9. Click **OK**.
10. Click **OK** again.

D4. Test the Set Action

1. Go to your dashboard.
2. Click a team in **Winner Distribution**.
3. Observe the chart.
4. The selected team should now be highlighted according to the set.

Purpose

This allows the user to interactively select a team instead of manually changing the set.

D5. Create a URL Action

1. Go to:

Dashboard → Actions

2. Click:

Add Action → Go to URL

3. Enter:

Open Team Information

4. Select:

Winner Distribution

under Source Sheets.

5. Set **Run action on to**:

Menu

6. In the URL box, enter:
<https://en.wikipedia.org/wiki/>

7. Click **Insert Field**.

8. Select:

Winner

The URL will become dynamic using the selected Winner.

9. Click **OK**.
10. Click **OK** again.

D6. Test the URL Action

1. Return to the dashboard.
2. Hover over a team.
3. Click the small ▼ menu.
4. Select **Open Team Information**.
5. Tableau will open the corresponding URL.

Edit Set Action

Name: Update Selected Team Set Insert ▾

Source Sheets: Dashboard 1 ▾

Run action on:
☒ Hover
☒ Select
☐ Menu
☐ Single-select only

Target Set: Selected Teams Set ▾

Running the action will:
☒ Assign values to set
☐ Add values to set
☐ Remove values from set

Clearing the selection will:
☐ Keep set values
☒ Add all values to set
☐ Remove all values from set

Buttons: Cancel OK

Winner Distribution

Winner	Count of Winner
Chennai Super Kings	2
Delhi Capitals	1
Gujarat Titans	1
Kolkata Knight Riders	1
Lucknow Super Giants	1

Top Venues

Venue	Count of Venue
Chepauk Stadium	2

Toss Decision Impact

Sub-Category	Count
Laptop	1
Rice	1

Add URL Action

Name: Open Team Information Insert ▾

Source Sheets: Dashboard 1 ▾

Run action on:
☒ Hover
☐ Select
☒ Menu

URL Target:
☒ New Tab if No Web Page Object Exists
☐ New Browser Tab

URL: https://en.wikipedia.org/wiki/Chennai_Super_Kings Insert ▾

Data Values:
Winner: Chennai Super Kings
Count of Winner: 2

Buttons: Cancel OK

Winner Distribution

Winner	Count of Winner
Chennai Super Kings	2
Delhi Capitals	1
Gujarat Titans	1
Kolkata Knight Riders	1
Lucknow Super Giants	1

Context Menu:
✓ Keep Only
Exclude
Open Team Information
In / Out of Selected Teams Set: In
Winner: Chennai Super Kings
Count of Winner: 2



Part E:-Create dynamic dashboards using parameters

This is the most important interactive part because it allows one chart to display different dimensions.

We will allow the user to choose:

- Winner
- City
- Venue
- Toss Decision

E1. Create the Parameter

1. Open **Winner Distribution**.
2. In the Data pane, click the small dropdown arrow.
3. Select:

Create Parameter

4. Set the name to:

Select Dimension

5. Set:

Data Type → String

6. Under **Allowable values**, select:

List

7. Enter:

Winner

City

Venue

Toss Decision

8. Click **OK**.

E2. Create Dynamic Dimension

1. Right-click in the Data pane.
2. Select:

Create Calculated Field

3. Name it:

Dynamic Dimension

4. Enter:

```
CASE [Select Dimension]
WHEN 'Winner' THEN [Winner]
WHEN 'City' THEN [City]
WHEN 'Venue' THEN [Venue]
WHEN 'Toss Decision' THEN [Toss Decision]
END
```

5. Click **OK**.

E3. Replace Winner with Dynamic Dimension

1. Open **Winner Distribution**.
2. Remove **Winner** from Rows.
3. Drag **Dynamic Dimension** to Rows.
4. Remove the old **Count of Winner** measure if necessary.
5. Drag **Number of Records** to Columns.

If **Number of Records** is not available, you can use another suitable count measure from your dataset.

E4. Display the Parameter

1. Find **Select Dimension** in the Data pane.
2. Right-click it.
3. Select:

Show Parameter

A parameter control will appear.

E5. Test the Parameter

Select:

Winner

The chart displays the winners.

City

The chart changes to cities.

Venue

The chart changes to venues.

Toss Decision

The chart changes to toss decisions.

Result

One visualization can now display different dimensions based on the user's selection.

E6. Add Parameter to Dashboard

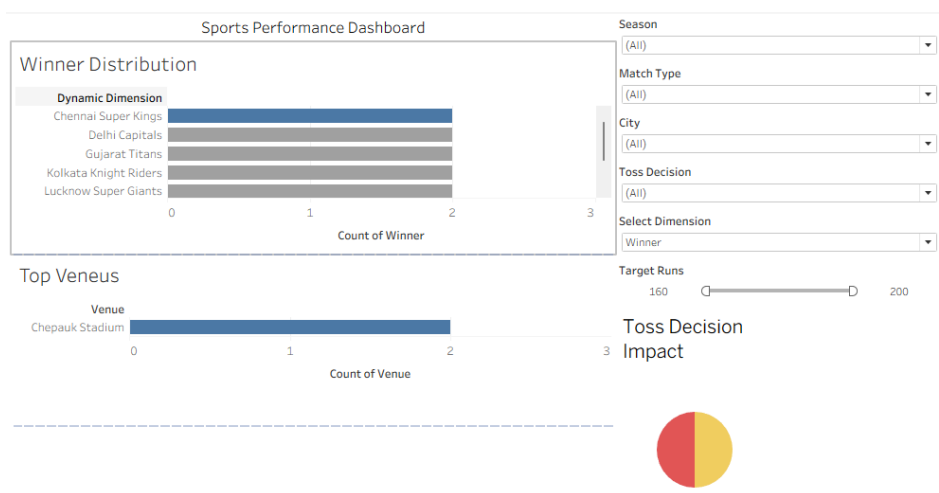
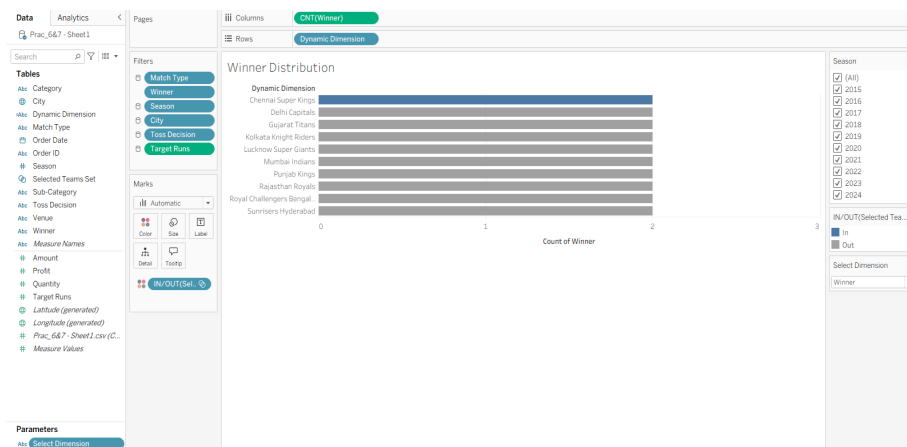
1. Open your dashboard.
2. Click the dropdown arrow on the **Winner Distribution** worksheet.
3. Select:

Parameters → Select Dimension

4. The parameter control will appear.
5. Position it near your other filters.

Your dashboard can now contain:

- Season
- Match Type
- City
- Toss Decision
- Target Runs
- Select Dimension



Part F:-Design dashboards for multiple devices

Tableau allows the same dashboard to be adjusted for different screen sizes.

F1. Desktop Layout

1. Open the dashboard.
2. Set the device to:

Desktop

3. Arrange your dashboard approximately like this:

IPL PERFORMANCE DASHBOARD

Season Match Type City

Toss Decision Target Runs Select Dimension

Winner Distribution Top Venues

Toss Decision Impact

4. Make sure all charts are readable.
5. Make sure filters are visible.
6. Remove unnecessary blank space.

F2. Tablet Layout

1. Open the **Device Preview** section.
2. Select:

Tablet

3. Create/customize the tablet layout.
4. Move the filters toward the top.
5. Arrange the charts into two sections.

For example:

Season Match Type

City Toss Decision

Target Runs Select Dimension

Winner Distribution

Top Venues

Toss Decision Impact

6. Resize the worksheets to fit the tablet screen.

F3. Phone Layout

1. Select:

Phone

2. Create/customize the phone layout.
3. Arrange everything vertically.

Recommended order:

IPL Performance Dashboard

Season

Match Type

City

Toss Decision

Target Runs

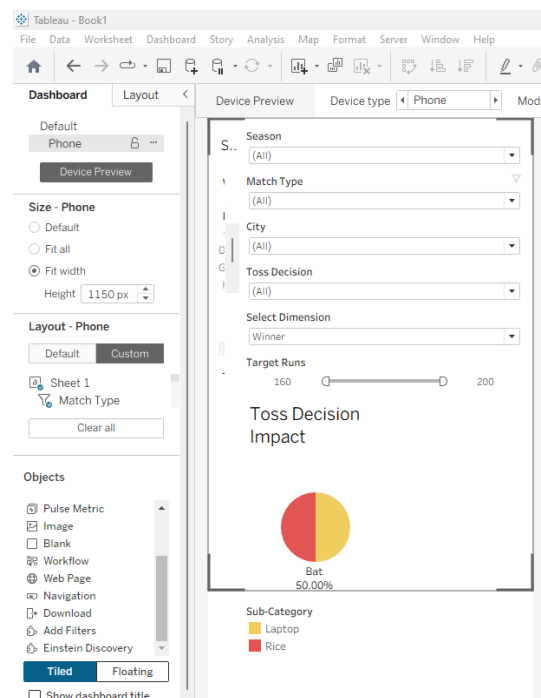
Select Dimension

Winner Distribution

Top Venues

Toss Decision Impact

4. Resize the charts.
5. Make sure text and filters are readable.
6. Make sure there is no overlap.



F4. Test All Device Layouts

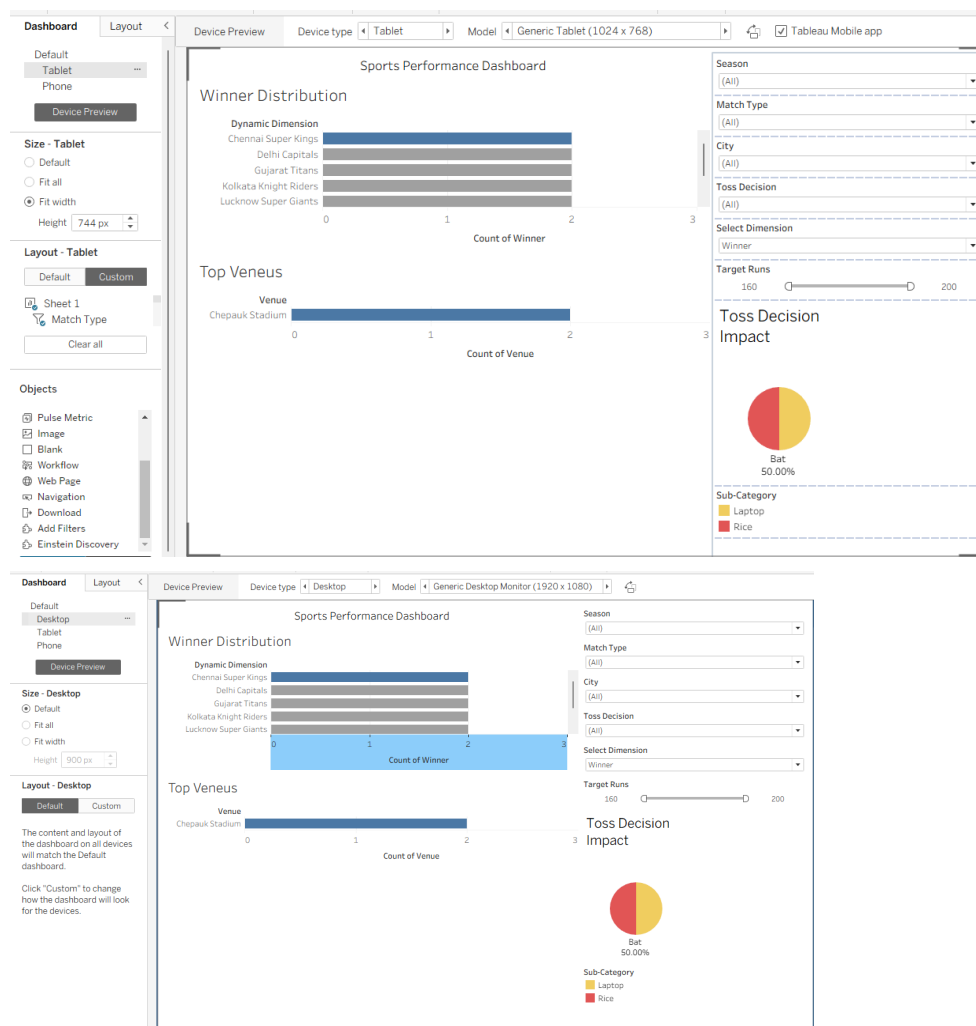
Check the dashboard in:

Desktop → Tablet → Phone

Test:

- Season
- Match Type
- City
- Toss Decision
- Target Runs
- Select Dimension
- Winner Set Action
- URL Action

Everything should remain functional.



Part G:- Build a complete data story using Story feature

G1. Create a New Story

1. At the bottom of Tableau, click:

New Story

2. Rename it:

IPL Performance and Match Analysis

3. Set the story size to:

Automatic**G2. Story Point 1 – Overall Match Performance**

1. Select the first story point.
2. Add your completed dashboard.
3. Set filters to **All**.
4. Give the story point the title:

Overall Match Performance

5. Add a description/caption explaining that the dashboard provides an overall view of:
 - Match winners
 - Top venues
 - Toss decisions
 - Match-related filtering

G3. Story Point 2 – Winner and Venue Analysis

1. Click:

New Blank Point

2. Add **Winner Distribution**.
3. Add **Top Venues** if appropriate.
4. Use the **Season** filter.
5. Select a season if you want to demonstrate a specific period.

Title:

Winner and Venue Analysis**What to observe**

Identify:

- Teams with more wins.
- Frequently occurring venues.
- Changes after applying the Season filter.

G4. Story Point 3 – Toss Decision Analysis

1. Click **New Blank Point**.
2. Add **Toss Decision Impact**.
3. Use the **Toss Decision** filter.
4. Change the Season filter if required.

Title:

Impact of Toss Decisions**What to observe**

The visualization allows you to examine the distribution of toss decisions and compare the results under different selections.

G5. Story Point 4 – Interactive Analysis

If your practical requires demonstrating interactivity, create another story point.

1. Click **New Blank Point**.
2. Add your interactive dashboard.
3. Demonstrate:
 - Set Action
 - URL Action
 - Parameter
 - Filters

Title:

Interactive Match Analysis

The user can select a team, change the dimension, and modify the filters to explore the dataset.

G6. Add an Annotation

1. Open the worksheet containing an important mark.
2. Right-click the mark.
3. Select:

Annotate → Mark

4. Enter an appropriate observation, such as:

This team has a high number of wins in the selected dataset.

5. Click **OK**.

The annotation will help explain the important finding.

G7. Final Story Structure

Your final Tableau Story can contain:

Story Point 1

Overall Match Performance

Complete dashboard overview.

Story Point 2

Winner and Venue Analysis

Winner and venue trends.

Story Point 3

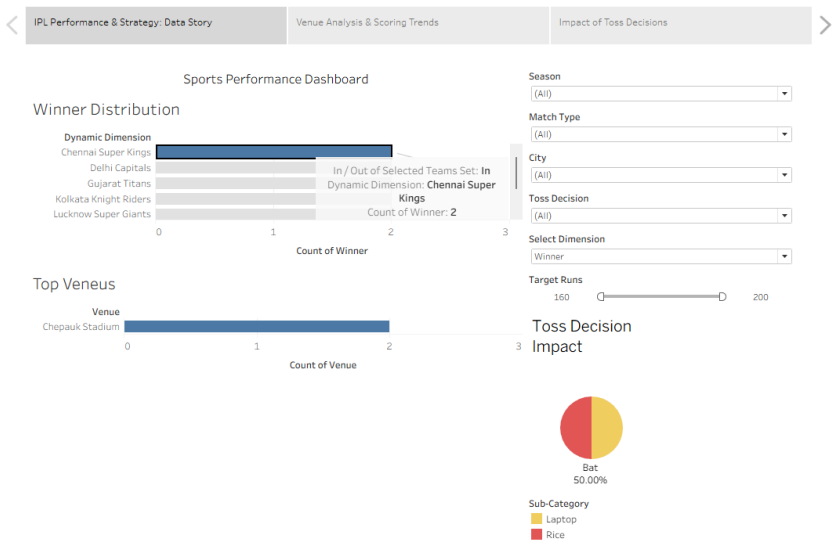
Impact of Toss Decisions

Toss decision analysis.

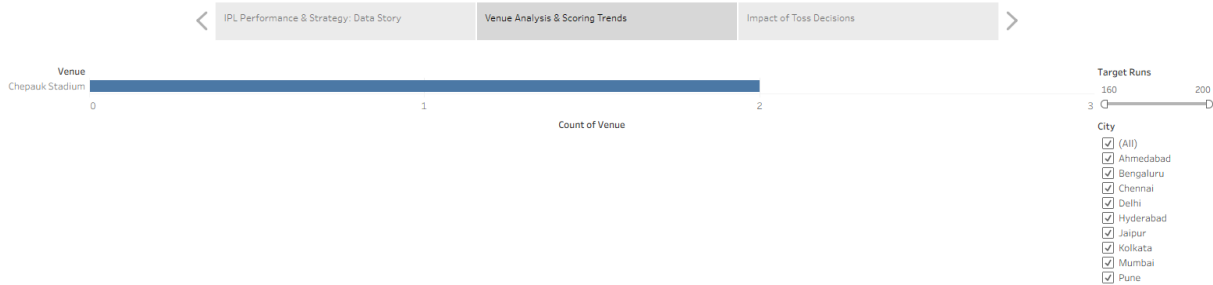
Story Point 4

Interactive Match Analysis

Filters, parameters, set action, and URL action.



Story 1



Story 1

